

PEC-4-8 : Machine Learning

Course Code	Course Name	Teaching Scheme (Weightage in Hr.)					Evaluation Scheme (Weightage in %)				
		L	T	P	S	Cr	Theory			Laboratory	
							MSE	TA	ESE	ISE	ESE
ME-xxxx	Machine Learning	3	0	0	1	3	30	20	50	--	

Course Outcomes:

Students who successfully complete this course will have demonstrated an ability to:

- CO1:** Explain the fundamental concepts, types, and applications of machine learning in engineering and interdisciplinary contexts.
- CO2:** Formulate supervised and unsupervised learning problems and apply appropriate algorithms for classification, regression, and clustering.
- CO3:** Analyze model performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrices.
- CO4:** Implement machine learning algorithms using programming tools and libraries such as Python, scikit-learn, and TensorFlow.
- CO5:** Evaluate the ethical, societal, and security implications of machine learning systems in real-world applications.
- CO6:** Design and optimize machine learning workflows for engineering problems involving data preprocessing, feature selection, and model tuning.

Syllabus:

Units	Contents	Hrs.
1	Introduction to AI & Machine Learning in Mechanical Engineering: History and evolution of AI and ML, Differences between AI, ML, and Data Science. Importance of ML in engineering, Overview of ML approaches: Supervised, Unsupervised, Reinforcement Learning Basic concepts: Reasoning, problem-solving, knowledge representation, planning, perception	6
2	Feature Engineering and Data Preprocessing: Importance of data quality and preprocessing, Techniques for feature extraction: Statistical features, Principal Component Analysis (PCA), Feature selection methods, Dimensionality reduction techniques	7
3	Supervised Learning Algorithms: Linear and logistic regression, Decision trees and random forests, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Model evaluation metrics: Accuracy, precision, recall, F1-score	6
4	Unsupervised Learning and Clustering: Clustering algorithms: K-Means, Hierarchical clustering, Dimensionality reduction: PCA, t-SNE, Anomaly detection techniques, Applications in mechanical systems: Fault detection, pattern recognition	7

5	Deep Learning and Neural Networks: Introduction to neural networks and deep learning, Convolutional Neural Networks (CNNs) for image data, Recurrent Neural Networks (RNNs) for time-series data, Applications in mechanical engineering: Predictive maintenance, image-based inspections	8
6	Advanced Topics and Applications in Mechanical Engineering: Reinforcement learning and its applications, Physics-informed machine learning models, Integration of ML with Computer-Aided Engineering (CAE) tools, Case studies: Digital twins, smart manufacturing, structural health monitoring	7

Suggested learning resources:

Textbooks:

1. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn and TensorFlow Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.
2. Introduction to Machine Learning with Python by Andreas C. Müller & Sarah Guido
3. Machine Learning For Dummies by John Paul Mueller & Luca Massaron

Reference Books:

1. Tom Mitchell "Machine Learning" McGraw Hill Publication, ISBN :0070428077 9780070428072
2. Marc Peter Deisenroth (Author), A. Aldo Faisal (Author), Cheng Soon Ong (Author), MATHEMATICS FOR MACHINE LEARNING, Cambridge University Press (23 April 2020); Cambridge University Press.

CO-PO Mapping

CO Code	Mapped POs	Description and Justification
CO1	PO1, PO2	Students apply mathematical and engineering fundamentals to understand machine learning principles and algorithms. They analyze problem domains and identify suitable learning paradigms, supporting foundational and analytical competencies.
CO2	PO2, PO3, PO4	Learners formulate supervised and unsupervised learning problems and select appropriate algorithms. They design and implement models for classification, regression, and clustering, integrating problem analysis, solution design, and research-based methods.
CO3	PO2, PO4	Students evaluate model performance using statistical metrics and validation techniques. This involves analytical thinking and interpretation of experimental results, reinforcing problem analysis and research-based evaluation.
CO4	PO5, PO12	Learners implement machine learning algorithms using modern programming tools and libraries. This fosters proficiency in contemporary technologies and promotes lifelong learning through hands-on experimentation and tool adaptation.
CO5	PO6, PO8	Students assess the ethical, societal, and security implications of machine learning systems. They reflect on responsible AI practices, data privacy, and fairness, supporting societal awareness and professional ethics.
CO6	PO3, PO5, PO12	Learners design and optimize machine learning workflows, including data preprocessing and model tuning. This outcome strengthens solution design, tool usage, and continuous learning in evolving data-driven environments.

CO-PO-PSO Mapping

Mapped Levels:

3 – High Contribution, 2 – Moderate Contribution, 1 – Low Contribution, 0– No Significant Contribution

CO Code	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	0	0	0	0	0	0	0	0	0	0	2	0	0
CO2	0	3	3	2	0	0	0	0	0	0	0	0	3	2	0
CO3	0	3	0	3	0	0	0	0	0	0	0	0	2	0	0
CO4	0	0	0	0	3	0	0	0	0	0	0	2	3	2	0
CO5	0	0	0	0	0	3	0	3	0	0	0	0	0	3	0
CO6	0	0	3	0	2	0	0	0	0	0	0	3	3	3	0

VI- 4.12 : Course: Deep Learning

Course Code	Course Name	Teaching Scheme (Weightage in Hr.)					Evaluation Scheme (Weightage in %)				
		L	T	P	S	Cr	Theory			Laboratory	
							MSE	TA	ESE	ISE	ESE
<td>	Deep Learning	3	1	0	0	4	30	20	50	--	--

Course Outcomes:

Students who successfully complete this course will have demonstrated an ability to:

- CO1:** Explain the foundational concepts of neural networks, activation functions, and learning algorithms used in deep learning.
- CO2:** Analyze the architecture and training dynamics of deep neural networks, including convolutional and recurrent models.
- CO3:** Apply optimization techniques and regularization strategies to improve model performance and generalization.
- CO4:** Design and implement deep learning models for classification, regression, and sequence modeling tasks using appropriate frameworks.
- CO5:** Evaluate model performance using metrics, validation techniques, and interpretability tools to ensure robustness and fairness.
- CO6:** Explore the role of deep learning in interdisciplinary applications such as computer vision, natural language processing, and healthcare analytics.

Syllabus:

Unit	Contents	Hrs.
1	Introduction to Deep Learning and Neural Networks: Overview of Machine Learning and Deep Learning, Biological vs. Artificial Neural Networks, Activation Functions and Loss Functions, Optimization Techniques: Gradient Descent, Backpropagation, Overfitting, Underfitting, and Regularization Methods	8
2	Convolutional Neural Networks (CNNs) for Mechanical Applications: Convolution Operations and Pooling Layers, Architectures: LeNet, AlexNet, VGG, ResNetTransfer Learning and Fine-Tuning, Applications: Defect Detection, Thermal Imaging Analysis	7
3	Recurrent Neural Networks (RNNs) and Time-Series Analysis: RNN Architectures: Vanilla RNNs, LSTM, GRU, Sequence Modeling and Prediction, Applications: Vibration Analysis, Predictive Maintenance, Challenges: Vanishing/Exploding Gradients, Sequence Length Handling	7
4	Autoencoders and Generative Models: Autoencoders: Structure and Training, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Applications: Design Optimization, Anomaly Detection	6

5	Reinforcement Learning in Mechanical Systems: Fundamentals of Reinforcement Learning (RL), Deep Q-Networks (DQN) and Policy Gradient Methods, Applications: Robotics Control, Adaptive Systems, Simulation Environments: OpenAI Gym, Custom Simulators	6
6	Integration and Deployment of Deep Learning Models: Model Deployment Strategies: Edge Computing, Cloud Services, Tools: TensorFlow Lite, ONNX, Docker, Case Studies: Real-world Applications in Mechanical Engineering Ethical Considerations and Model Interpretability	8

Suggested learning resources:

Textbooks:

1. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn and TensorFlow Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.
2. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press
3. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer Nature.
4. François Chollet, Deep Learning with Python, Manning; First Edition

Weblinks:

https://onlinecourses.nptel.ac.in/noc20_cs62/preview

CO-PO Mapping

Course Outcome (CO)	Mapped Program Outcomes (POs)	Description & Justification
CO1: Explain the foundational concepts of neural networks, activation functions, and learning algorithms used in deep learning.	PO1, PO2	This outcome builds theoretical understanding of deep learning principles, enabling students to apply mathematical and algorithmic reasoning to neural network behavior. It supports analytical thinking and foundational engineering knowledge.
CO2: Analyze the architecture and training dynamics of deep neural networks, including convolutional and recurrent models.	PO2, PO4, PO5	Students develop the ability to investigate and interpret the behavior of complex models, enhancing their skills in experimentation, performance evaluation, and use of modern tools.
CO3: Apply optimization techniques and regularization strategies to improve model performance and generalization.	PO2, PO3, PO5	This outcome strengthens students' problem-solving and design capabilities by integrating techniques that enhance model robustness and reduce overfitting, using both theoretical and practical approaches.

CO4: Design and implement deep learning models for classification, regression, and sequence modeling tasks using appropriate frameworks.	PO3, PO5, PO6	Students engage in end-to-end model development using industry-standard platforms, while considering ethical, safety, and sustainability aspects in deployment.
CO5: Evaluate model performance using metrics, validation techniques, and interpretability tools to ensure robustness and fairness.	PO4, PO5, PO11	This outcome promotes investigative rigor, tool proficiency, and responsible engineering practice by encouraging students to assess fairness, transparency, and reliability in model evaluation.
CO6: Explore the role of deep learning in interdisciplinary applications such as computer vision, natural language processing, and healthcare analytics.	PO6, PO7, PO12	Students examine the societal impact and interdisciplinary relevance of deep learning, fostering awareness of emerging technologies, ethical considerations, and lifelong learning.

CO-PO-PSO Mapping

Mapping levels

3 – Strongly related, **2** – Moderately related, **1** – Slightly related, **0** – Not related

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	0	0	0	0	0	0	0	0	0	0	3	1	0
CO2	2	3	1	3	2	0	0	0	0	0	0	0	3	2	0
CO3	2	2	2	2	3	0	0	0	0	0	0	0	3	3	0
CO4	1	2	3	2	3	1	1	0	0	0	0	0	3	3	0
CO5	1	1	2	3	3	0	0	0	1	0	2	0	2	3	0
CO6	1	1	0	1	1	3	2	1	0	0	1	3	2	3	0